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Question **1**

Complete

Not graded

Test the convergence of the series

$$\sum_{n=1}^{\infty} \frac{sin(2n-1)\pi}{n^{5/2}}.$$

Select one:

- a. The series converges conditinally.
- b. The series converges absolutely.
- c. The series diverges.

sin((2n-1)n) is zero for n = 1,2,3,.....

Therefore,

$\left|\frac{\sin((2n-1)\pi}{n^{5/2}}\right| = \frac{0}{n^{5/2}} = 0,$

As the absolute value of nth term is zero therefore,

$|\sum_{n=1}^{\infty}|a_n| = 0$

Therefore given series is converges absolutely

The correct answer is: The series converges absolutely.

Question **2**

Complete

Not graded

Test the convergence of the series

$$\sum_{n=1}^{\infty} \frac{n^2 + n}{n^4 + 3}.$$

Select one:

- a. The series converges conditionally.
- b. The series converges absolutely.
- c. The series diverges.

The nth term

$$\frac{n^2 + n}{n^4 + 3}$$

is positive for all n. Because

$$\frac{n^2 + n}{n^4 + 3} < \frac{n^2 + n^2}{n^4} < \frac{2}{n^2},$$

it follows that

$$\sum_{n=1}^{\infty} \frac{2}{n^2} = 2\sum_{n=1}^{\infty} \frac{1}{n^2}$$

is a convergent p-series for which $p = 2$.

Therefore, the original series converges absolutely.

The correct answer is: The series converges absolutely.

Question **3**

Complete

Not graded

Test the convergence of the series

$$\sum_{n=2}^{\infty} \frac{(-1)^n}{ln\, n}.$$

Select one:

- a. The series converges conditionally.
- b. The series converges absolutely.
- c. The series diverges.

The series

$$\sum_{n=2}^{\infty} \frac{(-1)^n}{\ln n} = \frac{1}{\ln 2} - \frac{1}{\ln 3} + \frac{1}{\ln 4} - \frac{1}{\ln 5} + \frac{1}{\ln 6} \cdots$$

is an alternating series. It is easy to verify that

$$a_n = \frac{1}{\ln n} \geq \frac{1}{\ln(n+1)} = a_{n+1}$$

for all $n \geq 2$, and $\lim_{n\to\infty} a_n = 0$.

Therefore, the alternating series applies.

The series converges conditionally.

The correct answer is: The series converges conditionally.

Question **4**

Complete

Not graded

Test the convergence of the series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2+1}.$$

Select one:

- a. The series converges conditionally.
- b. The series converges absolutely.
- c. The series diverges.

Take the absolute value of the nth term.

$$\left|\frac{(-1)^{n+1}}{n^2+1}\right| = \frac{1}{n^2+1}$$

Because $1/n^2+1$ is less than $\frac{1}{n^2}$, you know that this series converges by comparison with the p-series for which $p = 2$. Therefore, the original series converges absolutely.

The correct answer is: The series converges absolutely.

Question **5**

Complete

Not graded

Test the convergence of the series

$$\sum_{n=1}^{\infty} \frac{cosn\pi}{n}.$$

Select one:

- a. The series converges conditionally.
- b. The series converges absolutely.
- c. The series diverges.

$$\sum_{n=1}^{\infty} \frac{\cos n\pi}{n} = -\frac{1}{1} + \frac{1}{2} - \frac{1}{3} + \frac{1}{4} - \frac{1}{5} \cdots$$

$$= \sum_{n=1}^{\infty} \frac{(-1)^n}{n}$$

This is an alternating series with $a_n = \frac{1}{n}$, which decreases. Therefore, the original series converges conditionally by the alternating series test.

The correct answer is: The series converges conditionally.

Question **6**

Complete

Not graded

Test the convergence of the series

$$\sum_{n=1}^{\infty} \frac{1}{n^{2n}}.$$

Select one:

- a. The series converges conditionally.
- b. The series converges absolutely.
- c. The series diverges.

Because the nth term is positive for all

n and the terms of the series are less than or equal to those of a convergent p-series, for which

$p = 2$, $\sum_{n=1}^{\infty} \frac{1}{n^2}$, the original series

$\sum_{n=1}^{\infty} \frac{1}{n^{2n}}$ converges absolutely.

The correct answer is: The series converges absolutely.

Question **7**

Complete

Not graded

Test the convergence of the series

$$\sum_{n=1}^{\infty} \frac{(-1)^n 2^{2n}}{3^n}.$$

Select one:

- a. The series converges conditionally.
- b. The series converges absolutely.
- c. The series diverges.

$$\frac{(-1)^n 2^{2n}}{3^n} = \left(\frac{(-1)(2^2)}{3}\right)^n$$

$$= \left(\frac{-4}{3}\right)^n$$

Therefore,

$$\sum_{n=1}^{\infty} \frac{(-1)^n 2^{2n}}{3^n} = \sum_{n=1}^{\infty} \left(\frac{-4}{3}\right)^n.$$

This is a geometric series with

$|r| = \frac{4}{3} > 1$. Thus, the series diverges.

The correct answer is: The series diverges.

Question **8**

Complete

Not graded

Test the convergence of the series

$$\sum_{n=3}^{\infty} \frac{(-1)^n ln\, n}{n}.$$

Select one:

- a. The series converges conditionally.
- b. The series converges absolutely.
- c. The series diverges.

The series

$$\sum_{n=3}^{\infty} \frac{(-1)^n \ln n}{n}$$

is an alternating series. Use the derivative of

$f(x) = \frac{\ln x}{x}$ to show that

$$a_n = \frac{\ln n}{n} \geq \frac{\ln(n+1)}{n+1} = a_{n+1}$$

for all $n \geq 3$. Take the limit of a_n.

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{(-1)^n \ln n}{n} = 0.$$

Therefore, the alternating series test applies and the series converges conditionally.

The correct answer is: The series converges conditionally.

Question **9**

Complete

Not graded

Test the convergence of the series

$$\sum_{n=1}^{\infty}(-1)^n\left(\frac{1+n}{n}\right)^n.$$

Select one:

- a. The series converges conditionally.
- b. The series converges absolutely.
- c. The series diverges.

You know that

$$\lim_{n\to\infty}\left((-1)^n\left(\frac{n+1}{n}\right)^n\right)=\begin{cases}-e \text{ if } n \text{ is even}\\ e \text{ if } n \text{ is odd}\end{cases}$$

Because $\lim_{n\to\infty} a_n \neq 0$, the series diverges.

The correct answer is: The series diverges.

Question **10**

Complete

Not graded

Test the convergence of the series

$$\sum_{n=1}^{\infty} \frac{(-1)^n 2^{2n}}{7^n}.$$

Select one:

- a. The series converges conditionally.
- b. The series converges absolutely.
- c. The series diverges.

Take the absolute value of the nth term.

$$\left|\frac{(-1)^n 2^{2n}}{7^n}\right| = \left(\left|\frac{-4}{7}\right|\right)^n$$

$$= \left(\frac{4}{7}\right)^n.$$

Therefore, the series is a geometric series.

$$\sum_{n=1}^{\infty} \frac{(-1)^n 2^{2n}}{7^n} = \sum_{n=1}^{\infty} \left(\frac{4}{7}\right)^n,$$

with $|r| = \frac{4}{7} < 1.$

Thus, the series converges absolutely.

The correct answer is: The series converges absolutely.

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